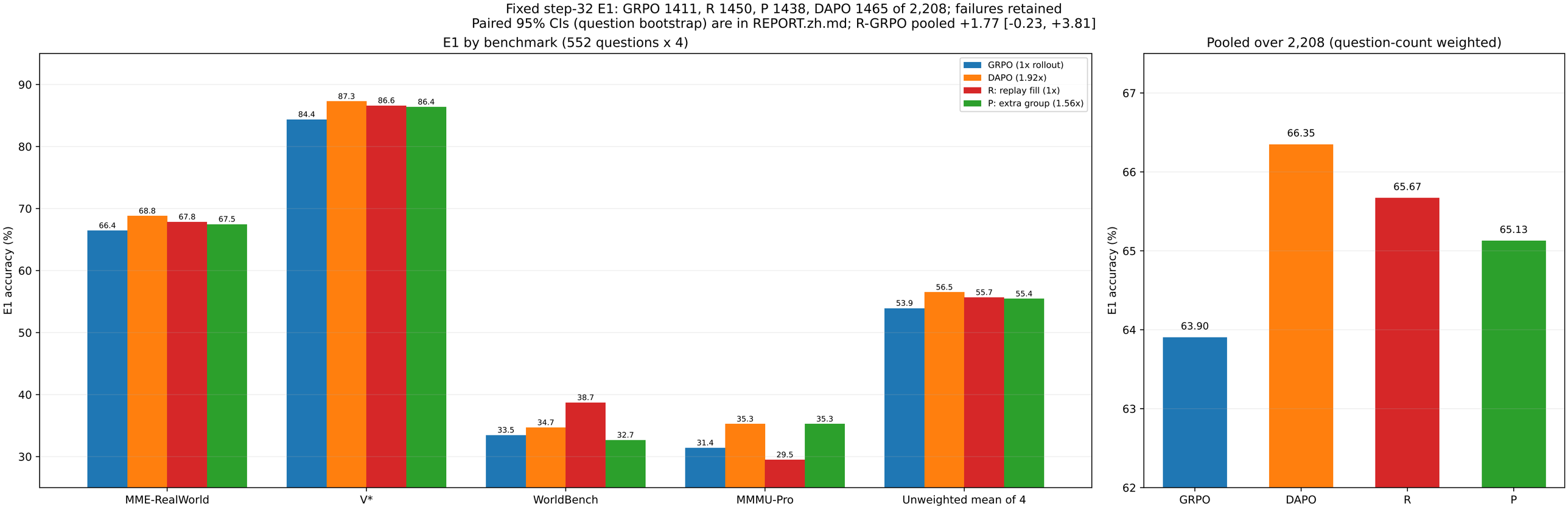
Fixed step-32 E1: GRPO 1411, R 1450, P 1438, DAPO 1465 of 2,208; failures retained
Paired 95% CIs (question bootstrap) are in REPORT.zh.md; R-GRPO pooled +1.77 [-0.23, +3.81]
E1 by benchmark (552 questions x 4)
GRPO (1x rollout)
DAPO (1.92x)
R: replay fill (1x)
P: extra group (1.56x)
E1 accuracy (%)
90
80
70
60
50
40
30
66.4
68.8
67.8
67.5
84.4
87.3
86.6
86.4
33.5
34.7
38.7
32.7
31.4
35.3
29.5
35.3
53.9
56.5
55.7
55.4
MME-RealWorld
V*
WorldBench
MMMU-Pro
Unweighted mean of 4
Pooled over 2,208 (question-count weighted)
E1 accuracy (%)
67
66
65
64
63
62
63.90
66.35
65.67
65.13
GRPO
DAPO
R
P